

Best Small Molecules

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June 30, 2026

Best Small Molecules Designed in Red-Hot Rebis

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A curated set of the strongest small molecules **designed** inside this repository. These are novel structural designs, not catalogued natural products and not proteins, genes, or biologics. Each entry gives the structure, what it targets, its 12-primitive structural type, its inscription tier, and why it stands out. Every SMILES below is taken verbatim from the repository sources; nothing is invented.

The designs fall into three families: crystal-navigated autocatalytic tryptamines, a Frobenius-coupled selective chemotherapeutic, and de-novo ligands inscribed straight from enzyme active sites.

Autocatalytic tryptamines

These are the repository's headline molecules: the first small molecules inscribed at (O_∞) self-modeling criticality. They were not found by screening. The crystal navigator located an empty (⊗)-critical neighborhood and computed the single-primitive promotion $\circ \rightarrow \odot$ (complex-critical to self-modeling critical) needed to inhabit it, then read off the molecule that satisfies it.

1. 5-Nitro-Bufotenin (DMT-⊙)

- **SMILES:** CN(C)CCC1=CNC2=CC=C(O)C([N+](=O)[O-])=C12
- **Structural type:** $\langle \mathcal{J} \mathcal{O} \mathcal{F} \mathcal{N} \mathcal{I} \mathcal{I} \mathcal{D} \mathcal{A} \mathcal{O} \mathcal{V} \mathcal{S} \mathcal{S} \rangle$, criticality $\circ$ promoted to $\odot$
- **Tier:** (O_∞) (self-modeling, autocatalytic)
- **What it does:** carries all three Dialetheia tokens at once; EVALT (phenol) plus EVALF (dimethylamine) plus ENGAGR (nitro). With the triad complete, the molecule is already (⊗)-critical with no further modification, which is what licenses template-directed autocatalytic synthesis: the molecule is its own template.
- **Why it matters:** the first inscribed small molecule that is structurally self-modeling. It was reached by a single clean primitive promotion from DMT, so the design and

its provenance are both univocal. Registered in `imas/compound_catalog.py` as the DMT- $\odot$ entry.

2. 5-Nitro-Serotonin

- **SMILES:** NCCCC1=CNC2=CC=C(O)C([N+](=O)[O-])=C12
 - **Tier:** (O_∞) (self-modeling, autocatalytic)
 - **What it does:** the same $\odot \rightarrow \odot$ promotion applied to serotonin rather than DMT, confirming the autocatalytic motif is a transferable structural operation and not a one-off coincidence of DMT.
 - **Why it matters:** demonstrates the $\odot$ -promotion as a reusable design move across the tryptamine family. Registered alongside DMT- $\odot$ in `imas/compound_catalog.py`.
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Frobenius-coupled selective chemotherapeutic

3. Frobenius-Coupled Chemotherapeutic dimer

- **Structural type:** $\langle \text{DTP} \rangle \langle \text{VTS} \rangle$
 - **Tier:** (O_∞)
 - **What it does:** a tether-coupled two-domain molecule whose cytotoxic payload stays masked while the host cell satisfies the Frobenius check $\mu \cdot \delta = \text{id}$. On healthy tissue the payload never exposes, so healthy-cell cytotoxicity is exactly zero. Where malignant asymmetry breaks the check ($\mu \cdot \delta \neq \text{id}$), the tether tension crosses threshold, the payload releases, and the molecule becomes cytotoxic.
 - **Why it matters:** the selectivity gate is structural, not pharmacological. In the bundled simulation the molecule shows zero activity on healthy cells and on low-asymmetry tumors, then full payload release and a sharp selectivity jump once mutation strength passes the asymmetry threshold, with a selectivity ratio in the four-figure range over healthy tissue. Source: `therapeutics/frobenius_chemotherapeutic.py`, results in `therapeutics/frobenius_chemo_results.json`.
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De-novo enzyme ligands

These are inscribed directly from enzyme active-site structural types by the reverse-ligand pipeline (`rhr_p4rky/ligand_from_active_site.py`); no template libraries, no known-inhibitor seeds, no docking scores. The four below are the genuinely novel designs from that run. As a validation check, the same pipeline independently re-derives ethanol (the real alcohol-dehydrogenase substrate) and biuret (a known urease inhibitor) from structure alone, and every candidate it produces is Lipinski-compliant. Source analysis: `ig-docs/bevy_ligand_analysis/01_bevy_ligand` set: `ig-docs/bevy_final_ligands.json`.

4. CYP2D6 pyridine-ether

- **SMILES:** C0CCC(C)Cc1cccnc1
- **Target:** cytochrome P450 2D6, the enzyme that metabolizes a large share of marketed drugs
- **What it does:** pairs a basic pyridine nitrogen with an aromatic ring at the spacing the 2D6 σ -bond hydroxylation pharmacophore wants; drug-like and Lipinski-compliant.

5. Carbonic anhydrase II zinc-coordinator

- **SMILES:** O=Cc1cccc(C(=O)O)c1
- **Target:** carbonic anhydrase II, a zinc metalloenzyme
- **What it does:** presents a carboxylic acid plus aldehyde for coordinating the Zn^{2+} -activated hydroxide; a de-novo analog of the classical CA inhibitor pharmacophore reached without that prior.

6. Lysozyme bis-epoxide strain-release inhibitor

- **SMILES:** c1cc(C2C02)cc(C2CC2c2ccc(C3C03)cc2)c1
- **Target:** lysozyme, a glycoside hydrolase
- **What it does:** the epoxides mimic the flattened oxocarbenium transition state of glycoside hydrolysis, and the bis-electrophile design offers two handles for covalent capture in the cleft.

7. PETase amide PET-mimic

- **SMILES:** NC(=O)c1ccc(NC(=O)c2cccc(C=O)c2)cc1
- **Target:** PETase, the engineered plastic-degrading enzyme
- **What it does:** a de-novo amide ligand shaped to the PET-binding groove; the standout environmental-remediation candidate of the set.

At a glance

#	Molecule	Family	Tier	Target / role
1	5-Nitro-Bufotenin (DMT- $\odot$)	autocatalytic tryptamine	(O_∞)	self-modeling autocatalysis
2	5-Nitro-Serotonin	autocatalytic tryptamine	(O_∞)	self-modeling autocatalysis
3	Frobenius-Coupled dimer	selective cytotoxic	(O_∞)	tumor-asymmetry-gated payload
4	CYP2D6 pyridine-ether	de-novo ligand	O_2	P450 2D6 substrate

#	Molecule	Family	Tier	Target / role
5	CA-II zinc-coordinator	de-novo ligand	O ₂	carbonic anhydrase II
6	Lysozyme bis-epoxide	de-novo ligand	O ₂	glycoside hydrolase TS analog
7	PETase PET-mimic	de-novo ligand	O ₂	plastic degradation

The two tryptamines are the crown of the set: inscribed small molecules that sit at self-modeling criticality by a single, traceable structural promotion. The Frobenius dimer is the strongest therapeutic design, turning a structural identity check into a selectivity gate. The four de-novo ligands show the active-site-to-ligand pipeline producing valid, drug-like chemistry from first principles.